- Noodles and dumplings are manufactured from flour-based doughs and shaped, then simmered or poached in liquid.

2.5.1. Milling

Cereal processing is complex. The principal procedure is milling, i.e., the grinding of the grain so that it can be easily cooked and rendered into attractive foodstuff. The steps involved in the process of milling are:

- Rice is passed through two stone rubber discs rotating at different speeds and by shearing action on the grain, the hull is pulled away.
- This is then milled in a machine called pearlor to remove coarse outer layers of bran and germ by the process of rubbing, resulting in unpolished milled rice.
- Unpolished rice is liable to develop rancidity and so it is next polished in a brush machine which removes the aleurone layer and yields polished rice.
- Sometimes the polished rice is further treated in a device known as trumbol to give a coating of sugar and talc to produce a brighter shine on the rice.

The percentage of losses of different nutrients during milling are: Protein 15 percent, Fat 82 percent, Thiamine 85 percent, Riboflavin 70 percent and Pyridoxine 50 percent. The degree of milling determines the amount of nutrients removed.

2.5.2. Parboiling

Parboiling is a process in which rice has been partially boiled in the husk. This makes rice easier to process by hand, boost its nutritional profile and change its texture.

1. Conventional process: This consists of the following steps:

- Steeping paddy in cold for 2 or 3 days in large cement tanks.
- Steaming of the soaked paddy for 5-10 minutes
- Drying in the sun.

2. Hot soaking process: This consists of the following steps:

- Soaking of paddy in water at 65-70° C for 3-4 hours.
- Draining of water and steaming of soaked paddy in the same vessel for 5-10 minutes
- Drying of the paddy in the sun or in mechanical driers.

Advantages of parboiling

- Dehusking of parboiled rice is easy.
- Milled parboiled rice has greater resistance to insects and fungus.
- Loss of nutrients due to the removal of husk and bran in milling are decreased.
- Loss of water soluble nutrients due to washing of rice is less in parboiled rice compared to raw rice.
- Parboiling improves digestability.

2.5.3. Malting of cereals

Malting is a controlled germination process, which activates the enzymes of the resting grain resulting in the conversion of cereal proteins and other macromolecules.

UNIT-02_023-041.indd 32 22-01-2020 12:45:54